Computer Vision

EMARO - European Master on Advanced Robotics

Robotics Engineering Master Degree

Image filtering

The following items are the steps that you have to do in this lab session:

1. Noise Filtering
   - Add Gaussian (standard deviation=20) and salt & pepper (density=20%) noise to the provided images:
     - display the image, the noisy images, and their histograms.
   - Remove the noise by using a moving average, a low-pass Gaussian filter and a median filter:
     - use two different spatial supports: 3x3 pixels and 7x7 pixels;
     - display the filters;
     - display the resulting images and their histograms.

2. Image Sharpening
   - Implement the sharpening filter:
     - use a spatial support of 7x7 pixels;
     - display the filter;
     - display the resulting images.